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Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization

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Technical Report

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To

BSSE 4th Year Exam Committee
Institute of Information Technology
University of Dhaka

Subject: Submission of Technical Report - *APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization*

Sir,

I am respectfully submitting the technical report for my Software Project Lab (SE 801 / SPL-03) titled “APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization.”

This report comprehensively covers the project's technical details, including design, implementation, and testing phases.

I have made every effort to ensure the quality and completeness of this work. However, I will be grateful for any valuable feedback or suggestions that may help improve the project further.

Thank you for your time and kind consideration.

Sincerely,

Habibur Rahman Mahin

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Supervisor's Signature

Acknowledgement

I am grateful to the Institute of Information Technology, University of Dhaka, for providing me with the opportunity to work on the project “APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization.” I would like to express my sincere gratitude to my supervisor, Mohd. Zulfiquar Hafiz, Professor, Institute of Information Technology, University of Dhaka, for her continuous guidance, valuable suggestions, and support throughout the development of this project. Their insights and feedback played an important role in shaping the direction and quality of this work. Lastly, I would like to thank my classmates for their cooperation, discussions, and helpful inputs during different phases of the project.

Abstract

APEX-Net is a web-based intelligent diagnostic system designed to automate the screening of chest X-ray images across fourteen thoracic diseases. The system integrates a deep learning model based on DenseNet121 enhanced with an Anatomical Prior Attention Module (APAM) to produce multi-label predictions alongside visual heatmaps that highlight regions of concern. The platform is developed to be accessible to general users, patients, and medical trainees, particularly in low-resource environments where expert radiological support is limited.

The system combines modern web technologies with advanced artificial intelligence to deliver a scalable, interpretable, and efficient diagnostic tool. This report presents the system's design, architecture, implementation, and evaluation in a structured and comprehensive manner.

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1. Introduction

1.1 Background

Thoracic diseases such as pneumonia, pneumothorax, and cardiomegaly remain among the leading causes of mortality worldwide. Chest X-ray imaging is one of the most widely used diagnostic tools for detecting such conditions due to its affordability and accessibility. However, accurate interpretation of X-rays requires trained radiologists, whose availability is limited in many regions.

Recent advances in deep learning have enabled automated image analysis systems capable of assisting in medical diagnosis. Despite these advancements, existing systems often lack interpretability, struggle with rare disease detection, and fail to incorporate anatomical context.

1.2 Problem Statement

As identified in the project proposal, current AI-based chest X-ray systems suffer from three major limitations:

1. **Imbalanced Detection**

Models tend to perform well on common diseases but fail to detect rare but critical conditions.

2. **Lack of Anatomical Awareness**

Most models treat the image uniformly without focusing on medically relevant regions.

3. **Lack of Interpretability**

Predictions are often presented without visual explanations, reducing clinical trust.

APEX-Net addresses these issues by incorporating anatomical priors, improving rare disease detection, and providing heatmap-based explanations.

1.3 Objectives

The primary objectives of APEX-Net are:

- To develop an anatomically-aware deep learning model for chest X-ray analysis
- To ensure reliable detection across both common and rare thoracic diseases
- To provide interpretable outputs using heatmaps
- To design a scalable and user-friendly web application
- To enable deployment in low-resource environments

1.4 Scope of the System

The system is intended for:

- General public health awareness
- Medical students and trainees
- Preliminary screening assistance

It is important to note that APEX-Net is **not a replacement for professional medical diagnosis**, but rather a supportive tool.

1.5 Timeline

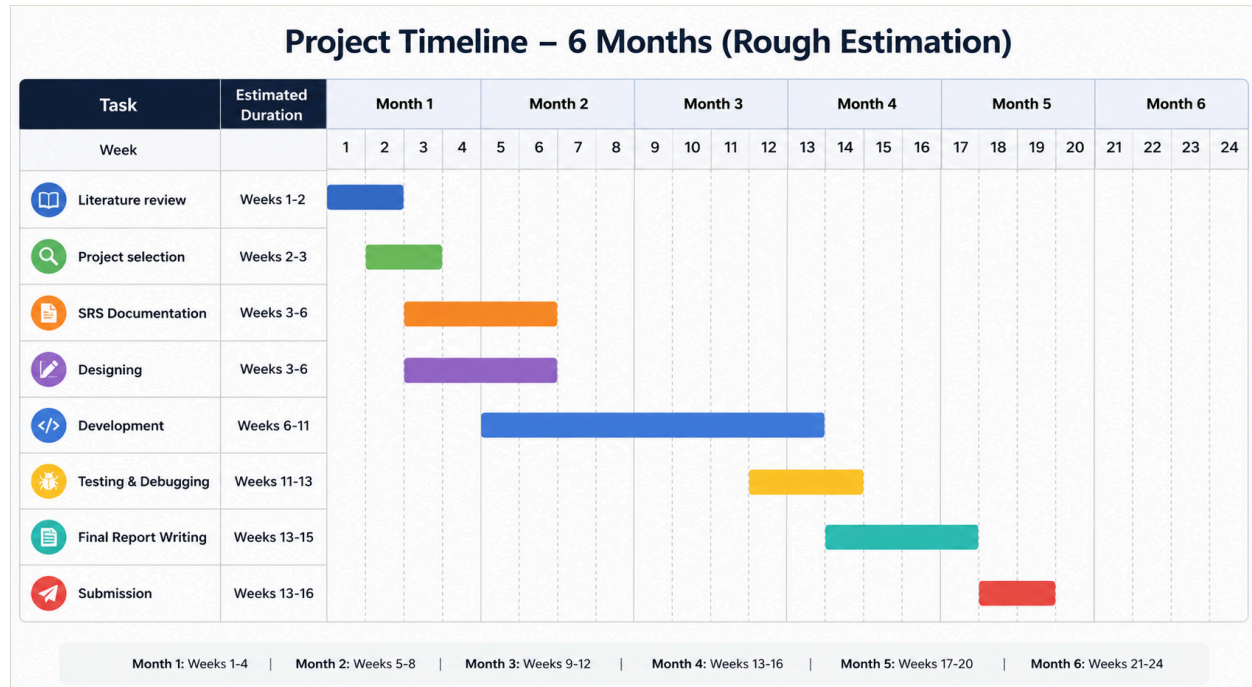


Figure-1 : Timeline of Project

2. Literature Review

Recent research in chest X-ray classification has focused heavily on deep convolutional neural networks. DenseNet-based architectures have shown strong performance due to their ability to reuse features and maintain gradient flow.

However, studies have highlighted limitations such as:

- Poor generalisation to rare conditions
- Lack of explainability
- Absence of anatomical reasoning

Attention-based models, such as those incorporating anatomical prior maps, have shown promise in addressing these issues. APEX-Net builds upon these advancements by integrating an Anatomical Prior Attention Module (APAM) into a DenseNet121 backbone.

3. System Overview

3.1 General Description

APEX-Net is a web-based system that enables users to upload chest X-ray images and receive automated diagnostic results. The system processes the image through a trained deep learning model and returns predictions along with a heatmap.

3.2 System Workflow

1. User registers or logs into the system
2. User uploads a chest X-ray image
3. Backend validates and preprocesses the image
4. Model performs inference
5. Heatmap is generated
6. Results are stored and displayed

3.3 Disease Categories

The system detects the following fourteen thoracic conditions:

Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, Pneumothorax

3.4 Output Interpretation

- Multi-label probability scores are generated
- If one disease probability is significantly higher, it is flagged as primary risk
- Heatmaps visually highlight affected regions

4. Elicitation of APEX-Net

The main task of the elicitation phase is to combine the elements of problem solving, negotiation, and specification. A collaborative working approach among stakeholders is essential to effectively gather and refine system requirements. For APEX-Net, multiple discussions and iterative feedback sessions were conducted to understand the needs of different user groups and ensure that the system design aligns with real-world expectations.

The following tasks have been completed for eliciting the requirements of APEX-Net:

- Collaborative Requirements Gathering
- Viewpoints of Stakeholders
- Quality Function Deployment
- User Story

4.1 Collaborative Requirements Gathering

During the inception phase of the APEX-Net project, several meetings and discussions were conducted with potential stakeholders to gather requirements. These sessions helped identify user expectations, system constraints, and functional priorities. Through continuous interaction and feedback, requirements were refined and negotiated to ensure feasibility and usability.

The stakeholders of APEX-Net include:

1. General Users (Patients / Public Users)
2. Medical Students and Trainee Doctors
3. Healthcare Professionals and Researchers

4.2 Viewpoints of Stakeholders

General Users (Patients / Public Users)

General users interact with APEX-Net as a diagnostic assistance tool. They expect a simple and intuitive interface where they can upload chest X-ray images and receive quick and understandable results. These users prioritise ease of use, fast response time, and clarity in output presentation.

Since most users may not have a medical background, the system must present predictions in a simplified manner. The inclusion of visual heatmaps helps users understand which areas of the X-ray are of concern. Additionally, users expect strong privacy and security measures to protect their medical data. The system serves as an initial screening tool that raises awareness and encourages timely medical consultation when necessary.

Medical Students and Trainee Doctors

Medical students and trainee doctors use APEX-Net as both a learning and diagnostic support tool. They are interested in understanding how artificial intelligence models interpret chest X-rays and identify different thoracic conditions.

The system provides probability distributions for multiple diseases, allowing students to analyse diagnostic uncertainty and compare different conditions. The heatmap visualisations further enhance their learning by showing region-specific importance, helping them correlate theoretical knowledge with practical image interpretation.

Access to historical scan data also enables them to study patterns and improve their clinical reasoning skills. Thus, APEX-Net acts as an educational platform that complements traditional medical training.

Healthcare Professionals and Researchers

Healthcare professionals and researchers focus on the reliability, consistency, and interpretability of the system. They expect accurate predictions across both common and rare diseases, as well as transparency in how results are generated.

The heatmap-based visual explanations are particularly valuable, as they provide insight into the model's decision-making process. This improves trust and allows professionals to validate the AI's output against clinical expectations.

Researchers may also use the system to analyse performance trends, evaluate model behaviour, and explore improvements in AI-based diagnostic systems. APEX-Net therefore serves as a foundational tool for further research and innovation in medical imaging.

4.3 QFD(Quality Function Deployment)

Quality Function Deployment (QFD) is a technique that translates the needs of the customers into technical requirements for software. Ultimately the goal of QFD is to translate subjective quality criteria into objective ones that can be quantified and measured, and can then be used to design and manufacture the product. It is a methodology that concentrates on maximizing customer satisfaction from the software engineering process. So, we have followed this methodology to identify the requirements for the project. The requirements which are given below, are identified successfully by the QFD.

4.3.1 Normal Features

Chest X-ray Upload and Analysis:

Users can upload chest X-ray images (JPG, PNG, or DICOM) through a simple web interface. The system processes the image and performs automated analysis using the trained deep learning model.

Multi-label Disease Prediction:

The system predicts probabilities across 14 thoracic diseases simultaneously. It highlights the most dominant condition when one disease probability is significantly higher.

Heatmap Visualization (Explainability):

For each prediction, the system generates a heatmap overlay that highlights the regions of the X-ray influencing the diagnosis, improving interpretability and user understanding.

Scan History Management:

Users can view previously uploaded scans along with timestamps, predictions, and heatmaps. This allows tracking and comparison of results over time.

4.3.2 Expected Features

Registration and Authentication:

Users register and log in using email and password. Authentication ensures secure access to personal scan history and protects user data.

Image Preprocessing and Validation:

Uploaded images are automatically validated and preprocessed (resizing, normalization) before being passed to the model, ensuring consistent input quality.

Asynchronous Inference Processing:

For handling larger workloads or slower inference, the system supports asynchronous processing to maintain responsiveness and scalability.

4.3.3 Exciting Features

Anatomically-Aware AI Model (APAM Integration):

The model focuses on anatomically relevant regions of the chest, improving detection accuracy by mimicking how radiologists interpret X-rays.

Rare Disease Sensitivity Optimization:

The system is designed to detect both common and rare thoracic conditions effectively through balanced training and per-disease tuning.

User-Friendly Diagnostic Interface:

Results are presented in a clear and structured format, combining probability scores with visual explanations, making the system accessible even to non-expert users.

4.4 User Story

APEX-Net aims to provide an accessible and intelligent solution for early detection of thoracic diseases through AI-powered chest X-ray analysis. By combining automated diagnosis with visual explanations, the system helps both medical professionals and general users understand results more clearly and supports decision-making in low-resource environments.

Upload and AI-Based Diagnosis

Users can upload a chest X-ray image to the platform and receive instant AI-powered analysis across 14 thoracic conditions. The system processes the image and provides predictions quickly, making it useful as a first-pass screening tool, especially in areas with limited access to radiologists.

Visual Heatmap Explanation

For each diagnosis, the system generates a visual heatmap highlighting the regions of concern in the X-ray. This allows users—including non-medical users—to understand where abnormalities are detected, improving interpretability and trust in the results.

Smart Anatomical Focus

The AI model focuses on anatomically relevant regions of the chest instead of analysing the entire image uniformly. This mimics how radiologists interpret X-rays and improves diagnostic accuracy by prioritising important areas.

Rare Disease Detection

The system is designed to detect not only common thoracic conditions but also rare and serious diseases. A customised training approach ensures balanced performance across all conditions, addressing limitations of traditional models.

Clinical Reliability and Accuracy

APEX-Net applies advanced training techniques and per-disease sensitivity tuning to ensure consistent and clinically meaningful results across all 14 conditions, improving reliability in real-world usage.

Lightweight Web Application

The platform is delivered as a lightweight web application, making it accessible and deployable in low-resource clinical environments. Users can access the system easily without requiring specialised hardware or software.

5. Scenario-based Modeling

The success of a computer-based system or product is measured in many ways, but user satisfaction resides at the top of the list. Understanding how actors want to interact with a system will lead the software team to properly characterize requirements and build meaningful analysis and design models. Hence, requirements modeling begins with the creation of scenarios in the form of use cases, activity diagrams, and swimlane diagrams.

5.1 What is a Use Case Diagram?

A use case is a written description of how users will perform tasks on this website. It outlines, from a user's point of view, a system's behavior as it responds to a request. Each use case is represented as a sequence of simple steps, beginning with a user's goal and ending when that goal is fulfilled. Use cases specify the expected behavior (what), and not the exact method of making it happen (how). Use cases, once specified, can be

denoted both in textual and visual representations (i.e., use case diagram). A key concept of use case modeling is that it helps us design a system from the end user's perspective. It is an effective technique for communicating system behavior in the user's terms by specifying all externally visible system behavior.

Primary Actor

Primary actors interact to achieve the required system function and derive the intended benefit from the system. They work directly and frequently with the software.

Secondary Actor

Secondary actors support the system so that primary actors can do their work. They either produce or consume information.

5.2 The Purpose of Use Case Diagram

The reasons why an organization would want to use case diagrams include:

- Represent the goals of systems and users.
- Specify the context a system should be viewed.
- Specify system requirements.
- Provide a model for the flow of events when it comes to user interactions.
- Provide an outside view of a system.
- Shows external and internal influences on a system.

5.3 Use Case Diagram: APEX-Net

5.3.1 Level 0:

Name: APEX-Net System

Primary actors: Users (General Users, Medical Students, Clinicians/Doctors)

Secondary actors: Central Database (PostgreSQL), Mail (Email Service), AI Model

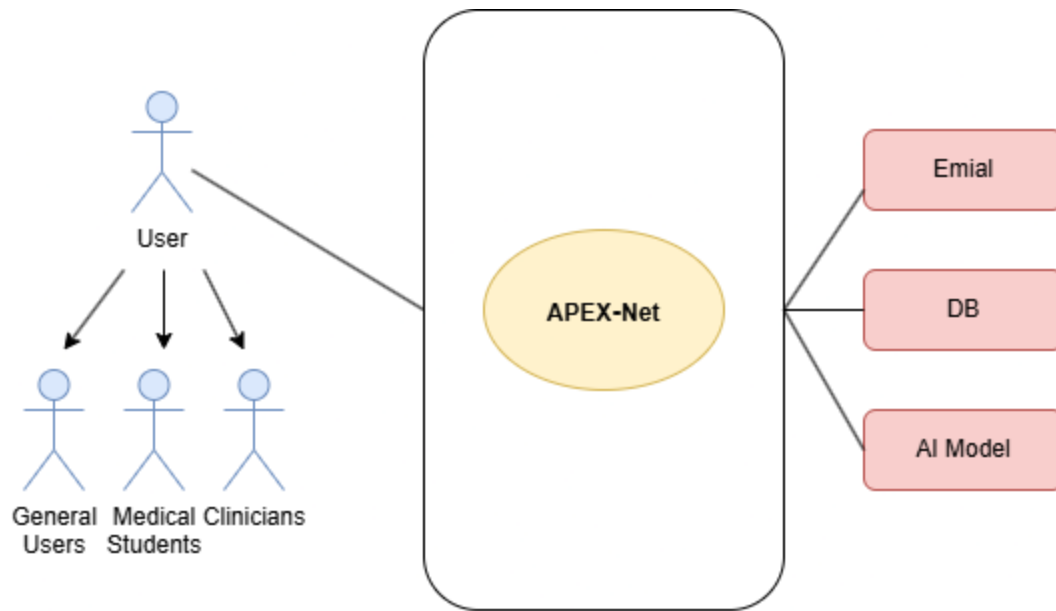


Figure-2 : Level 0 (APEX-Net system)

5.3.2 Level 1:

Name: APEX-Net System details

Primary actors: Users (General Users, Medical Students, Clinicians/Doctors)

Secondary actors: Central Database (PostgreSQL), Mail (Email), AI Model

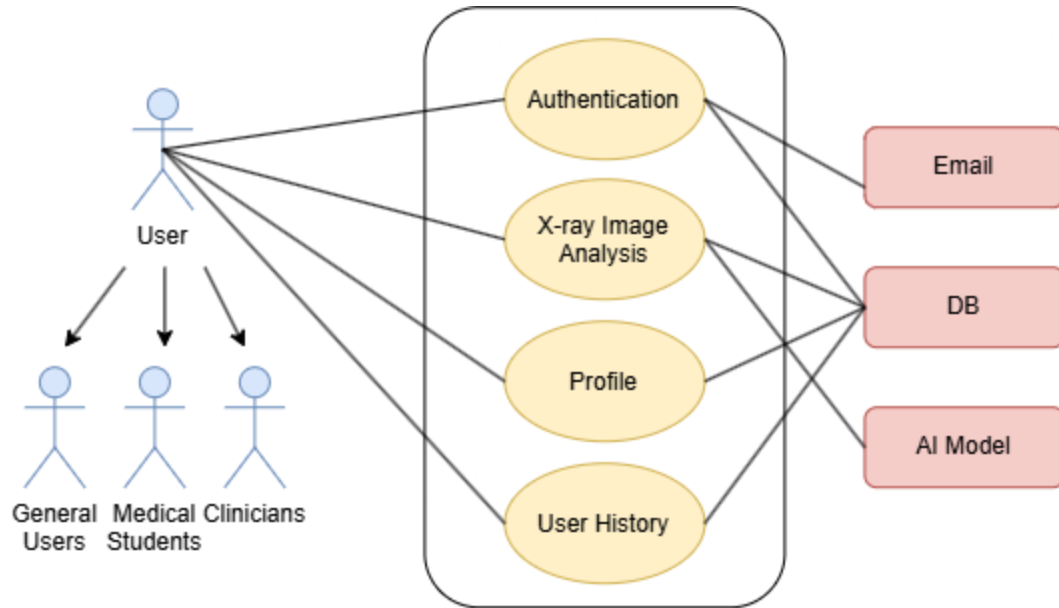


Figure-3 : Level 1 (APEX-Net System details)

5.3.3 Level 1.1:

Name: Authentication

Primary actors: Users (General Users, Medical Students, Clinicians/Doctors)

Secondary actors: Central Database (PostgreSQL), Mail (Email)

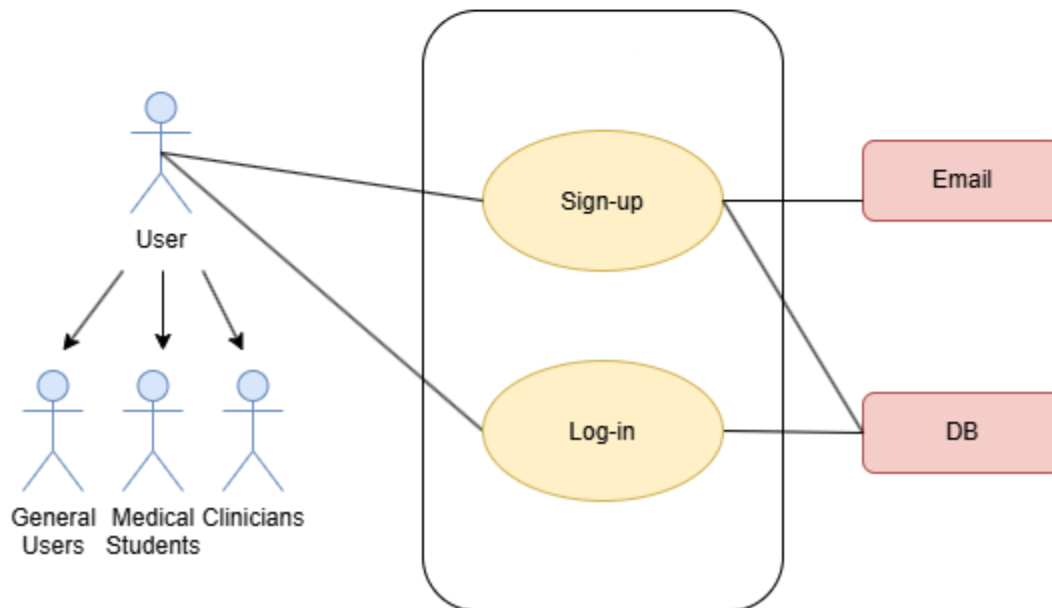


Figure-4 : Level 1.1 (User Signup & Login)

5.3.4 Level 1.2:

Name: X-ray Image Analysis

Primary actors: Users (General Users, Medical Students, Clinicians/Doctors)

Secondary actors: Central Database (PostgreSQL), AI Model

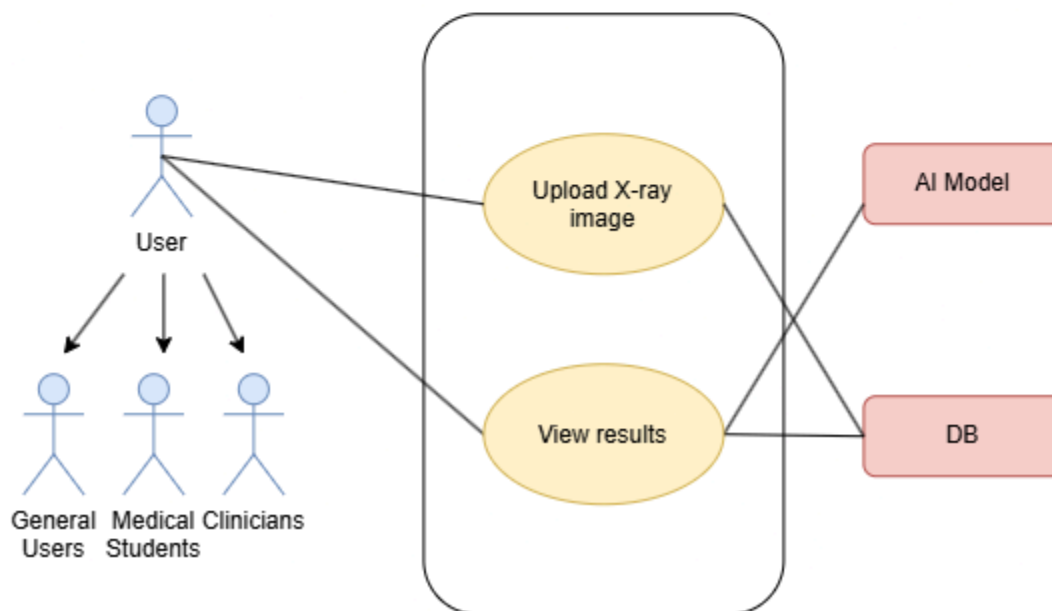


Figure-5 : Level 1.2 (X-ray Image Analysis)

5.3.5 Level 1.3:

Name: Update Profile

Primary actors: Users (General Users, Medical Students, Clinicians/Doctors)

Secondary actors: Central Database (PostgreSQL)

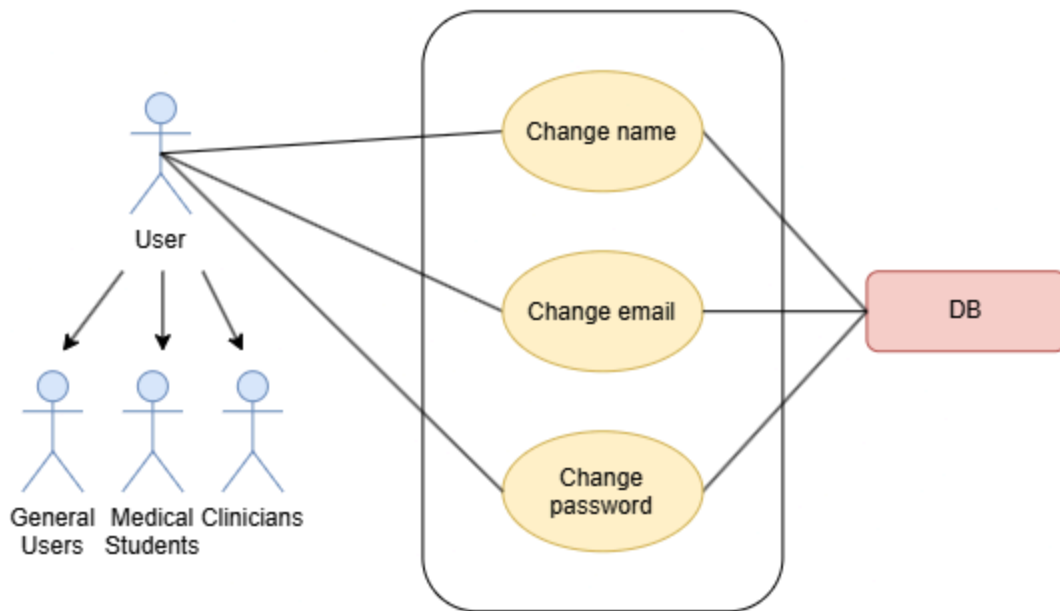


Figure-6 : Level 1.3 (Update Profile)

6. Activity Diagrams

Level :1

Name: APEX-Net System detailed

Reference: Use Case Diagram level - 1

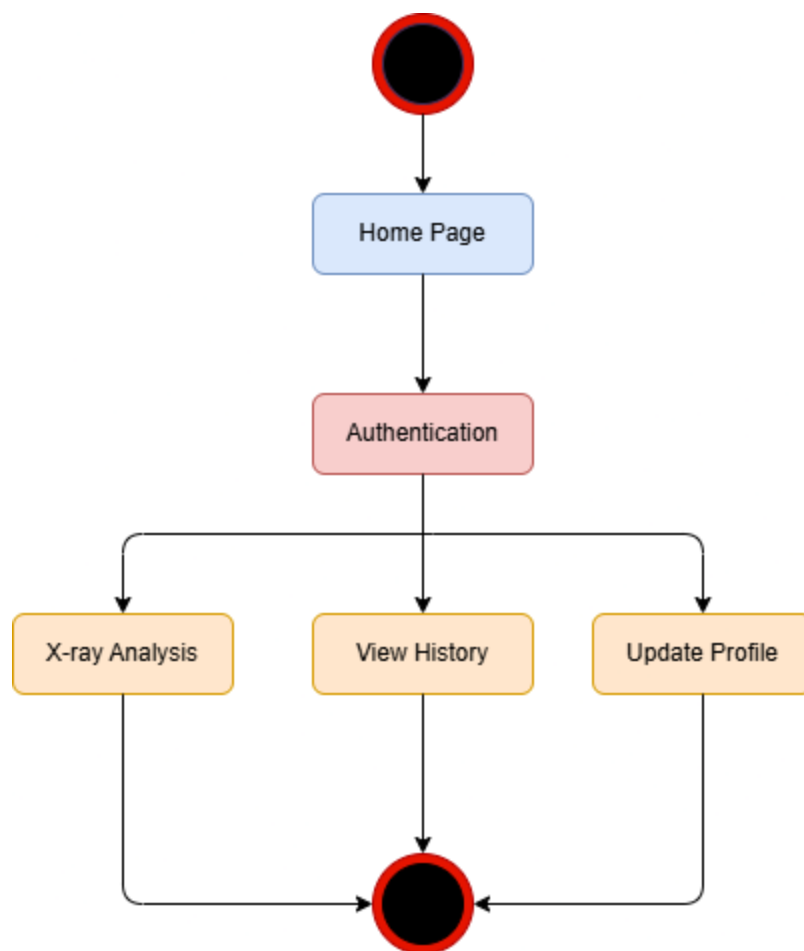


Figure-7 : Activity Diagram Level 1

Level :1.1

Name: Authentication

Reference: Use Case Diagram level - 1.1

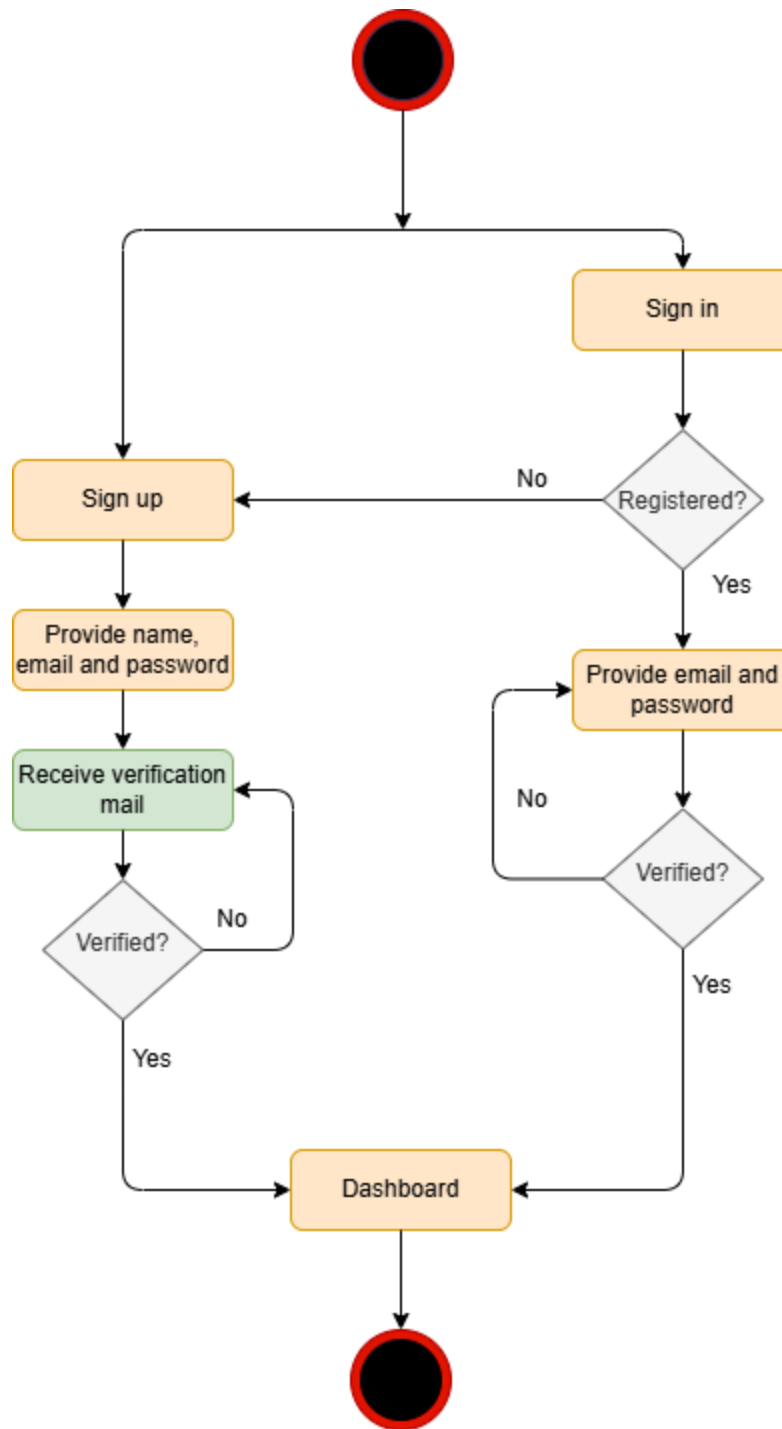


Figure-8 : Activity Diagram Level 1.1

Level :1.2**Name:** X-ray Image Analysis**Reference:** Use Case Diagram level - 1.2

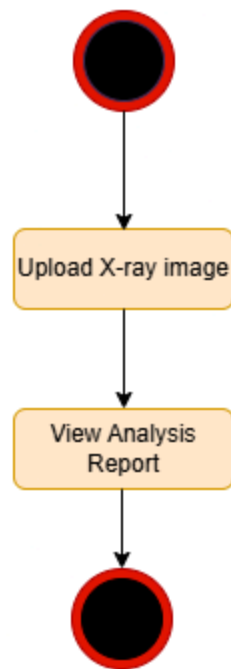


Figure-9 : Activity Diagram Level 1.2

Level :1.3

Name:Update Profile

Reference: Use Case Diagram level - 1.3

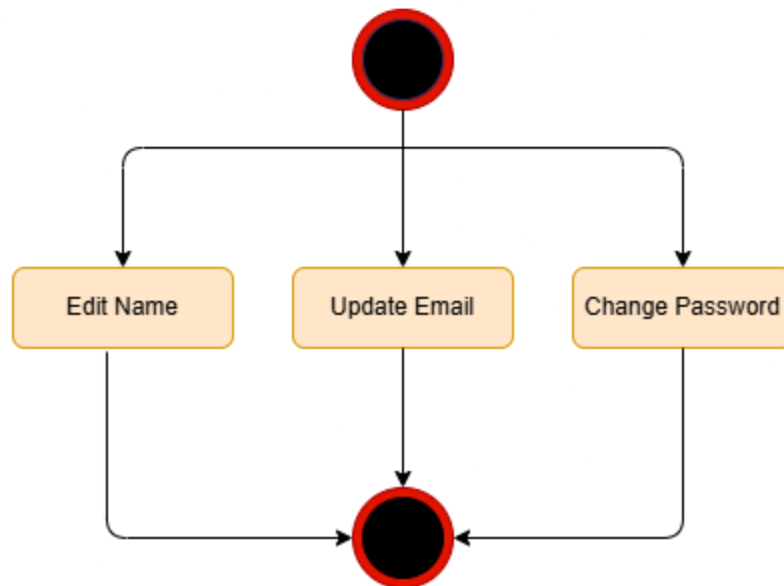


Figure-10 : Activity Diagram Level 1.3

7. Database Modeling

Data based modeling is a visual representation of a database structure and how data is organized within it. It involves creating entity-relationship diagrams (ERDs) that depict tables (entities), attributes (columns), and the relationships between them. Database modeling is typically achieved by analyzing the data requirements of a system, identifying entities and their attributes, and then using modeling tools or software to create ERDs that serve as a blueprint for database design and implementation.

7.1 Schemas of APEX-Net

Table	Attribute Name	Data Type
Profiles	id (PK)	UUID
	full_name	TEXT
	created_at	TIMESTAMPTZ
Xray_Uploads	id (PK)	UUID
	user_id (FK)	UUID
	file_name	TEXT
	file_size	INT4
	storage_path	TEXT
	uploaded_at	TIMESTAMPTZ
	created_at	TIMESTAMPTZ
Predictions	id (PK)	UUID
	user_id (FK)	UUID

	xray_upload_id (FK)	UUID
	prediction_data	JSONB
	confidence_score	FLOAT8
	analyzed_at	TIMESTAMPTZ
	created_at	TIMESTAMPTZ

7.2 ER Diagram

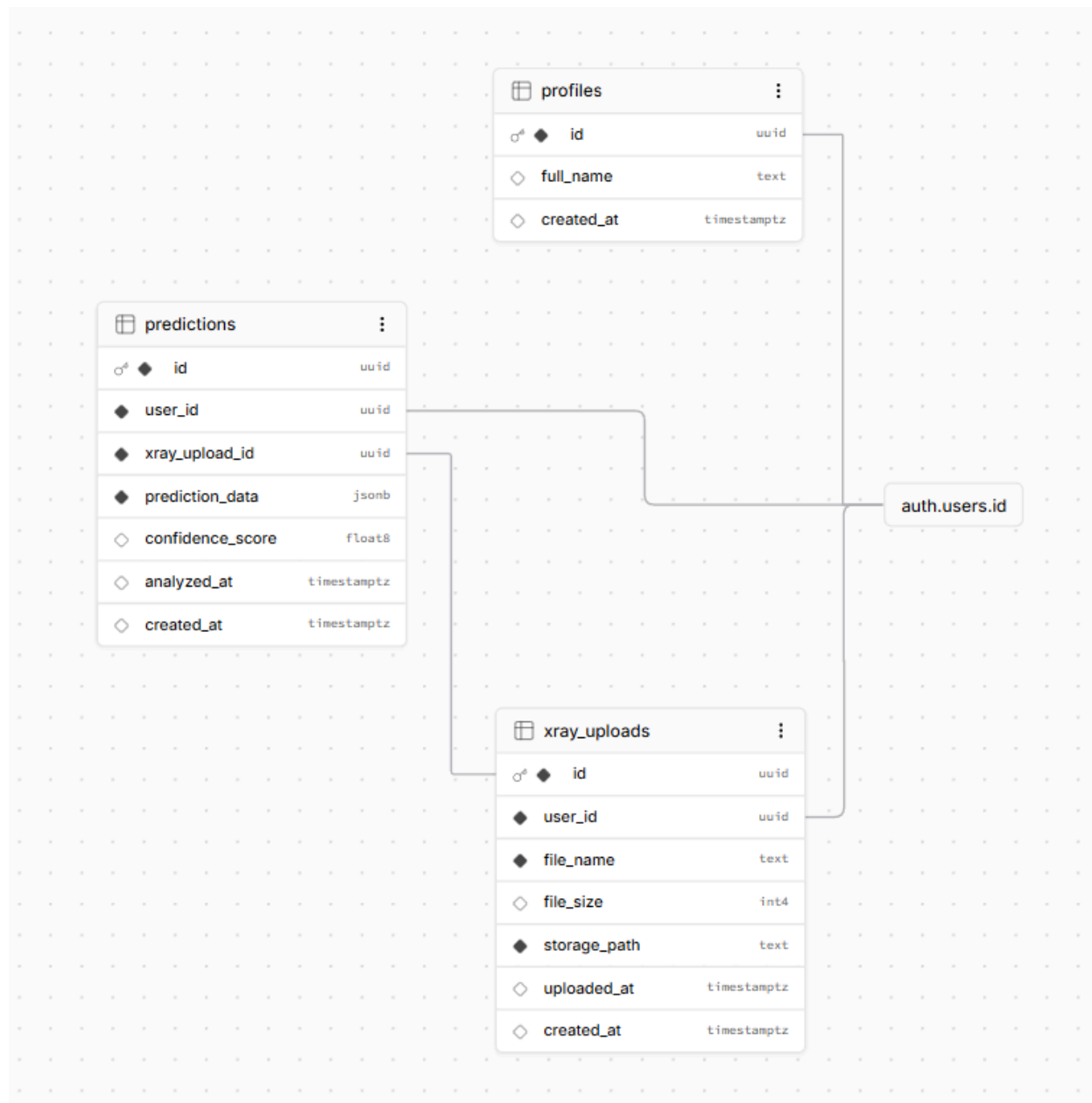


Figure-11 : ER Diagram of APEX-Net

8. Preliminary Test Plan

In this chapter, a high-level description of testing goals and a summary of items and features to be tested for APEX-Net are presented.

8.1 High-Level Description of Testing Goals

Serving as a bridge between users and the development team, software testing ensures the system is reliable, accurate, and user-friendly. For APEX-Net, the primary goal is to validate the correctness of AI predictions, system stability, and usability of the web application.

The goals of high-level software testing are given below:

- Ensure valid chest X-ray images are uploaded and processed correctly.
- Verify AI model generates accurate multi-label disease predictions.
- Confirm heatmap visualization correctly highlights relevant regions.
- Ensure invalid inputs are handled gracefully.
- Evaluate system performance under multiple concurrent uploads.
- Validate responsiveness and usability of the web interface.

8.2 Item Pass/Fail Criteria

If the system's output matches the expected result, the test case is considered successful. Otherwise, it is marked as failed.

Test Cases:

The following table provides the test cases that need to be executed for client-side and system testing:

Test ID	Test Cases	Input Test Data	Steps to be Executed	Expected Result
T1	Valid X-ray Upload	Chest X-ray (.jpg/jpeg/.png/.dcm)	i) Upload valid image ii) Confirm upload	Image accepted and ready for analysis
T2	Invalid Input Handling	Unsupported file (.txt/.pdf)	i) Upload invalid file	Error message: "Unsupported file format"
T3	AI Prediction Output	Valid chest X-ray	i) Upload image ii) Run analysis	Multi-label disease predictions displayed
T4	Heatmap Visualization	Valid chest X-ray	i) Upload image ii) Run analysis	Heatmap displayed highlighting affected regions
T5	UI Workflow	Multiple uploads	i) Upload image ii) View result iii) Navigate history	Smooth workflow; results displayed correctly
T6	Performance Under Load	Batch of X-ray images	i) Upload multiple images simultaneously	System processes all without crash; acceptable response time
T7	Prediction Consistency	Same image (multiple runs)	i) Upload same image multiple times	Similar predictions generated with minimal variation
T8	Data Storage Verification	Valid upload + prediction	i) Upload image ii) Run analysis	Data stored correctly in database (image + prediction results)

This preliminary test plan ensures that APEX-Net is tested for functionality, reliability, performance, and usability before final deployment.

9. Conclusion

This report presents the design and technical foundation of **APEX-Net**, an AI-powered web application for thoracic disease detection using chest X-rays. The system's objectives, requirements, and workflows have been clearly defined, along with its architecture and data structure.

APEX-Net integrates deep learning with a user-friendly interface to provide multi-label disease predictions along with visual heatmaps, improving interpretability and usability. Its lightweight, web-based design makes it suitable for deployment in low-resource settings, where access to expert radiologists is limited.

Overall, APEX-Net serves as an effective tool for preliminary screening and supports better understanding of medical imaging through explainable AI.

Future Development Scope:

- Integration of more advanced deep learning models to improve diagnostic accuracy
- Support for additional medical imaging modalities such as CT scans
- Real-time optimisation for faster inference and response
- Deployment as a mobile or cross-platform application
- Incorporation of clinician feedback for continuous model improvement
- Enhanced explainability techniques beyond heatmaps
- Integration with hospital or clinical information systems

In conclusion, APEX-Net demonstrates a practical and scalable approach to making AI-assisted medical diagnosis more accessible, interpretable, and reliable.

[The End]